

MASTER PRODUCT REQUIREMENTS DOCUMENT

REAL-TIME SOCIAL CHAT & MULTIPLAYER GAMES

Beginner-Safe • Full-Stack • Claude Handoff Ready

Version 3.0 • Exactly 10 Development Phases

Fixed Stack: Next.js + TypeScript + GitHub + Vercel + Supabase

This is the Master Source of Truth. It explicitly defines where the code, live website, database, authentication, profile pictures, realtime services, backend logic and secrets belong so a developer or Claude does not have to guess.

1. FIXED PLATFORM ARCHITECTURE

Item	Official platform	Purpose
Code	GitHub	Source code and version history.
Web app	Next.js + TypeScript	Frontend and application framework.
Deployment	Vercel	Live website plus server-side Next.js functions.
Backend/data platform	Supabase	PostgreSQL, Auth, Storage and Realtime.
Database	Supabase PostgreSQL	Permanent application data.
Authentication	Supabase Auth	Signup, login, logout and sessions.
File storage	Supabase Storage	Profile pictures.
Realtime	Supabase Realtime	Chat, presence, invitations and games.
Server logic	Next.js Route Handlers on Vercel	Secure validation and orchestration.
Custom domain	Optional later	Not required for MVP.

Critical: Supabase is the backend/data platform. Vercel is the website deployment platform. The database is Supabase PostgreSQL.

2. BEGINNER PLATFORM MAP

Thing	Simple meaning	This project uses it for
GitHub	Online code storage	All source code
Vercel	Makes website live	Deployment
Supabase	Backend platform	Database + Auth + Storage + Realtime
PostgreSQL	Actual database	Users, chats, games and history
Supabase Auth	Login system	Signup/login/logout
Supabase Storage	Cloud files	Profile pictures
Supabase Realtime	Instant updates	Chat, presence and games
Next.js	Web framework	Website + server-side routes

3. PRODUCT & SCOPE

Product: Real-Time Social Chat & Multiplayer Games Platform.

Registered users search one another by unique username, view profiles, privately chat in real time, send game invitations and play multiplayer games in protected rooms.

Required MVP

- Signup/login/logout; unique username; profile picture; name/bio; online/offline status.
- Username search such as @Ali with Message and Play Game.
- Private text/emoji chat with time, sent/delivered/read, typing, online status, history and unread count.
- Realtime game invitation with Accept/Decline and protected private rooms.
- Tic-Tac-Toe, Rock Paper Scissors, Connect Four and Memory Game.
- Dashboard: Search Users, Recent Chats, Online Users, Available Games, My Profile, Total Users and Online Now.
- Real frontend + backend + database + realtime; no UI-only fake implementation.
- Responsive mobile, tablet and desktop.

Non-goals

- Group chat.
- Public social feed.
- Voice/video calls.
- Payments.
- Advanced matchmaking/rankings/tournaments.
- Unrequested AI features.

4. USER JOURNEY & ROUTES

- Open Vercel website → Signup/Login → Search @username → Profile → Message → Realtime chat → Play Game → Invitation → Accept/Decline → Protected room → Realtime moves → Result/history → Dashboard.

Route	Purpose
/	Landing/auth
/signup	Signup
/login	Login
/dashboard	Dashboard
/users/[username]	Profile
/chat/[conversationId]	Private chat
/games	Game selection
/game/[roomId]	Live game
/profile	Own profile
/404	Fallback

5. SUPABASE DATABASE DESIGN

Supabase PostgreSQL is the permanent source of truth. Browser localStorage must not replace it.

Table	Core fields / purpose
profiles	id, username, display_name, avatar_url, bio, created_at, updated_at
conversations	id, participant_a, participant_b, created_at, updated_at
messages	id, conversation_id, sender_id, body, created_at, delivered_at, read_at
game_invitations	id, sender_id, recipient_id, game_type, status, created_at, responded_at
game_rooms	id, game_type, player_a, player_b, status, state, current_turn, winner, created_at, ended_at
game_moves	id, room_id, player_id, sequence, action, created_at
game_history	id, room_id, game_type, result, completed_at

- Username is unique and normalized.
- Conversation is exactly two users in MVP.
- Messages are ordered deterministically.
- Game moves use sequence numbers.
- Private records are protected by Supabase Row Level Security (RLS).
- Statistics are real database queries, never hard-coded.

6. AUTH, STORAGE, REALTIME & SECURITY

Supabase Auth

- Use Supabase Auth for signup/login/logout/session.
- Authenticated Supabase user ID is the identity source.
- Protect private routes.
- Never expose the service-role key.

Supabase Storage

- Use an avatar bucket for profile pictures.
- Restrict upload/update by storage policies.
- Store safe path/URL in profiles.
- Validate file type and size.

Supabase Realtime

- Messages, presence, typing, invitations, game state and game moves.
- Reconnect must resubscribe and synchronize persistent state.
- Realtime is synchronization; PostgreSQL remains persistent source of truth.

RLS

- Only profile owner can edit their profile.
- Only conversation participants can read messages.
- Only authorized game participants can access rooms.
- Validate every game action server-side.
- Never trust client winner/turn/player/room claims.

7. BACKEND & DEPLOYMENT

Layer	Location	Responsibility
Frontend	Next.js on Vercel	UI and navigation.
Server-side logic	Next.js Route Handlers/server functions on Vercel	Secure operations, validation and Supabase calls.
Backend services	Supabase	PostgreSQL, Auth, Storage, Realtime, RLS.
Source control	GitHub	Code and version history.

Deployment

- Connect GitHub repository to Vercel.
- Vercel builds Next.js and publishes preview/production deployments.
- Production environment variables are configured in Vercel.
- The live website uses the Vercel URL unless a custom domain is added later.
- Supabase is not the website host in this architecture.

Deployment flow: Code → GitHub → Vercel → Live Website → Supabase.

Secrets

- Never commit .env files or real keys.
- Keep .env.example with names only.
- Supabase service-role key is server-only.
- Private keys live in Vercel server environment variables.

8. CHAT, INVITATIONS & GAMES

Chat

- Text + emoji.

- Server timestamp.
- Sent/Delivered/Read.
- Typing indicator.
- Online/offline.
- Unread count.
- Persistent history.
- Realtime without refresh.
- Safe reconnect without duplicates.

Invitations

- Choose game → create invitation → realtime notification → Accept/Decline → accepted invitation creates protected room.

Games

- Tic-Tac-Toe: alternating turns, legal cells, three-in-row win, full-board draw.
- Rock Paper Scissors: choices hidden until both submit; server resolves.
- Connect Four: legal columns, gravity, four connected wins, draw.
- Memory Game: shared authoritative board, server validates selections, synchronized completion.

9. UI/UX & TESTING

- Modern, clean, professional design.
- Responsive mobile/tablet/desktop.
- Clear loading, empty, error and reconnect states.
- Accessible labels/focus states.
- Two independent authenticated sessions must be used for realtime QA.
- Test auth, profiles, search, chat, typing, presence, delivery/read/unread, invitations, all games, illegal actions, reconnect, RLS and deployment.

10. EXACTLY 10 DEVELOPMENT PHASES

Claude must follow these ten phases in order. No extra implementation phase and no uncontrolled one-shot build.

PHASE 01 — Foundation + GitHub + Supabase + Vercel

- Next.js + TypeScript.
- GitHub repository.
- Supabase project.
- Auth/database foundation.
- Storage/Realtime setup.
- Local environment.
- GitHub → Vercel.
- Preview deployment.
- Safe environment-variable structure.
- Architecture README.

Acceptance Gate: Next.js, GitHub, Supabase and Vercel are connected; preview is live; no secrets committed.

PHASE 02 — Authentication + Profiles

- Supabase Auth.
- Protected routes.
- Profiles table/RLS.
- Unique username.
- Avatar Storage.

- Name/bio.
- Profile editing.
- Public profile.

Acceptance Gate: Two real accounts can authenticate, edit profiles and upload avatars; unauthorized edits fail.

PHASE 03 — Dashboard + Search + Presence

- Dashboard.
- @username search.
- Profile result.
- Message/Play Game.
- Recent chats foundation.
- Online users.
- Real Total Users/Online Now.
- Realtime presence.

Acceptance Gate: User A finds B; actions work; dashboard data is real; presence works.

PHASE 04 — Persistent Private Chat

- Conversation table/RLS.
- Messages table/RLS.
- Private conversation.
- Text/emoji.
- History.
- Timestamps.
- Recent chats.
- Unread foundation.
- Indexes.

Acceptance Gate: Messages persist across reload and unauthorized reads fail.

PHASE 05 — Realtime Chat + Message States

- Realtime messages.
- Typing.
- Presence.
- Delivered/read.
- Unread.
- Reconnect.
- Resubscribe.
- Duplicate-event protection.

Acceptance Gate: Two sessions chat without refresh; states work; reconnect is safe.

PHASE 06 — Game Invitations + Private Rooms

- Game picker.
- Invitation state.
- Realtime invitation.
- Accept/Decline.
- Room creation.
- Authorization.
- Waiting state.
- Chat/game navigation.

Acceptance Gate: Invitation works in realtime; accepted invite creates protected room.

PHASE 07 — Tic-Tac-Toe + Rock Paper Scissors

- Server validation.
- Turn/cell/win/draw.
- Hidden RPS choices.
- Server result.
- Realtime state.
- History.
- Two-session QA.

Acceptance Gate: Both games fully work; invalid actions rejected; results persist.

PHASE 08 — Connect Four + Memory Game

- Connect Four validation.
- Memory authoritative state.
- Selection validation.
- Realtime state.
- History.
- Reconnect.
- Two-session QA.

Acceptance Gate: All four games work in realtime and persist results.

PHASE 09 — Integration + Security + Responsive Polish

- End-to-end integration.
- RLS/authorization audit.
- Input validation.
- Realtime listener audit.
- Mobile/tablet/desktop.
- Performance review.
- Remove mock core behavior.
- Vercel environment verification.

Acceptance Gate: Full journey passes; security is acceptable; responsive UI works; data is truly Supabase-backed.

PHASE 10 — Final QA + Production Deployment + Handover

- Full checklist.
- Two-user QA.
- Cross-device QA.
- Reconnect QA.
- Security/RLS QA.
- Production Supabase verification.
- Vercel production deployment.
- Final GitHub release.
- README/architecture/deployment docs.
- Final acceptance.

Acceptance Gate: All P0 requirements pass; Vercel site is live with production Supabase; documentation is complete.

11. CLAUDE RULES

- Treat this PDF as the Master Source of Truth.
- Fixed stack: Next.js + TypeScript + GitHub + Vercel + Supabase.
- Database = Supabase PostgreSQL.

- Deployment = Vercel.
- Auth = Supabase Auth.
- Storage = Supabase Storage.
- Realtime = Supabase Realtime.
- Do not switch to Firebase or another database/deployment platform without explicit human approval.
- Do not replace real backend/database/realtime with fake UI state.
- Do not invent major features.
- Inspect repository before editing.
- Implement one phase at a time.
- Run build/lint/tests after each phase.
- Use two independent sessions for realtime tests.
- Report changed files, tests, failures and acceptance status.
- Never expose service-role key.

12. CLAUDE MASTER INSTRUCTION

Build the complete functional application described in this PRD. Use Next.js + TypeScript, GitHub, Vercel and Supabase. Deploy the website on Vercel. Use Supabase PostgreSQL for persistent data, Supabase Auth for authentication, Supabase Storage for profile pictures and Supabase Realtime for realtime synchronization. Use secure Next.js server-side Route Handlers/functions on Vercel for server-side operations. Use RLS and server-side authorization. Never expose the service-role key. Implement exactly the ten phases in order. Do not invent major features or change the platform stack without explicit human approval. Test realtime with two independent authenticated sessions. Do not claim completion unless the phase acceptance gate passes.

13. FINAL DATA OWNERSHIP

Data/Function	System of record
Source code	GitHub
Live website	Vercel
Login/session	Supabase Auth
Profiles/database	Supabase PostgreSQL
Profile pictures	Supabase Storage
Messages/read/delivery/unread	Supabase PostgreSQL
Realtime chat/presence	Supabase Realtime + persisted DB state
Invitations	Supabase PostgreSQL + Realtime
Game rooms/moves/history	Supabase PostgreSQL + Realtime
Secure server-side logic	Next.js server functions on Vercel

14. FINAL ACCEPTANCE TEST

- Open Vercel URL.
- Create two accounts.
- Login in separate sessions.
- Search @username.
- Open profile.
- Chat with text/emoji.
- Verify realtime, typing, presence, delivered/read/unread.
- Send/accept game invitation.

- Enter protected room.
- Play all four games.
- Verify synchronized moves.
- Try invalid action and verify rejection.
- Finish and verify history in Supabase.
- Reconnect and verify recovery.
- Verify no secret is exposed.
- Verify correct Supabase tables/storage contain production data.

15. BEGINNER CHECKLIST

- Code → GitHub.
- Live website/deployment → Vercel.
- Database → Supabase PostgreSQL.
- Signup/Login → Supabase Auth.
- Profile pictures → Supabase Storage.
- Realtime → Supabase Realtime.
- Secure server-side app logic → Next.js server functions on Vercel.
- Private server secrets → Vercel server environment variables.
- Development order → Exactly 10 phases.

MASTER PRD v3.0 — SUPABASE + VERCEL — EXACTLY 10 PHASES — CLAUDE HANDOFF READY